Environment Management (EM)

This case study explains an integrated Environment Management process in detail. It, thus, fosters a thorough understanding of each process step and underlying SAP functionality to support sustainable enterprises.

Product

SAP S/4HANA 2023
Global Bike

SAP Fiori 3.0

Level

Beginner

Focus

Sustainability
Environment Management

Author

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Version

4.3.1

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MOTIVATION

In the environment management exercises you got familiar with locations, compliance requirements and compliance scenarios. All of these already existed in the system.

In this case study you will create a new location and a completely new scenario for that location.

PREREQUISITES

Before you use this case study, you should be familiar with navigation in the SAP system.

To successfully work through this case study, it is not necessary to have finished the EM exercises. However, it is recommended.

NOTES

This case study uses the model company Global Bike.

ACKNOWLEDGMENT

This case study was developed with the help of students and staff, as well as research and student assistants from SAP UCC Magdeburg.

SAP University Alliances

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Process Overview

Learning Objective Understand and perform the process of collecting and analyzing environmental data, of calculating sustainability measures such as the carbon footprint and of preparing standard ESG reports following given compliance requirements.

Time 140 min

Scenario Your colleague and you are tasked to create a location for the new packing station at Global Bike's distribution center in Miami, Florida. After that, you build and specify a compliance scenario for that location. In the process, you collect and load data, use calculations and aggregations for electricity consumption of that packing station. Additionally, you prepare environmental reports.

Employees involved Jae Kim, Plant Manager Miami, DC Miami

Alberto Conti, Technical Office Assistant, DC Miami

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Step 1: Create New Location

Task Create a new location. **Time** 10 min

Short Description Use the *Manage Locations* app to create a new location.

Name (Position) Jae Kim, Plant Manager Miami, DC Miami

Definition

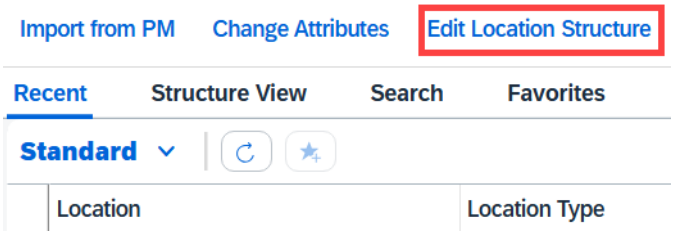
As a first step you need to create a new location. In SAP S/4HANA Environment Management (EM), a location can represent an identified real place, a virtual space, as well as an existing or planned object with relevance to sustainability. The location is a central master data object that can be organized in a structure to represent where a location exists spatially or functionally in a company. Examples of a location are a plant, a piece of equipment, or a work area.

Start

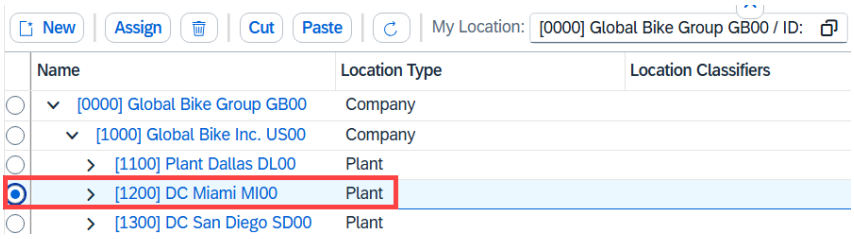
In the space *Environment Management* use the app *Manage Locations* to create a new location.



In the *Manage Locations* screen, click on Edit Location Structure .



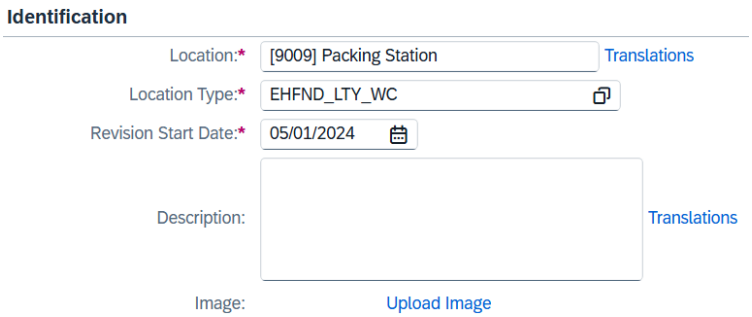
Now you see Global Bike's current location structure. In order to create a sub location you need to choose a location. Please select location *[1200] Plant Miami MI00* by clicking on the circle to the left of this location entry. You may have to explode the hierarchical location structure.



Now click New to create a new location in the Miami plant.

The *Create Location* screen appears. In the *Location field*, type **[9###] Packing Station** (please replace ### with your individual user ID). For *Location Type* use the value help icon and select **Work Center** and for *Revision Start Date* choose **05/01/2024** (May 1, 2024).

[9###] Packing Station
Work Center
05/01/2024

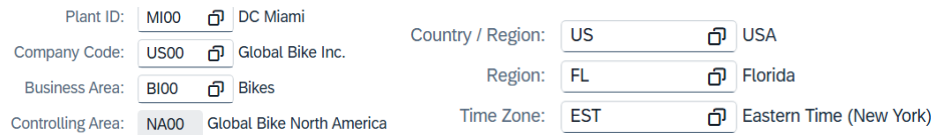


In the *Organizational Information* area, find and select *Plant ID* **MI00**, *Company Code* **US00** and *Business Area* **BI00**.

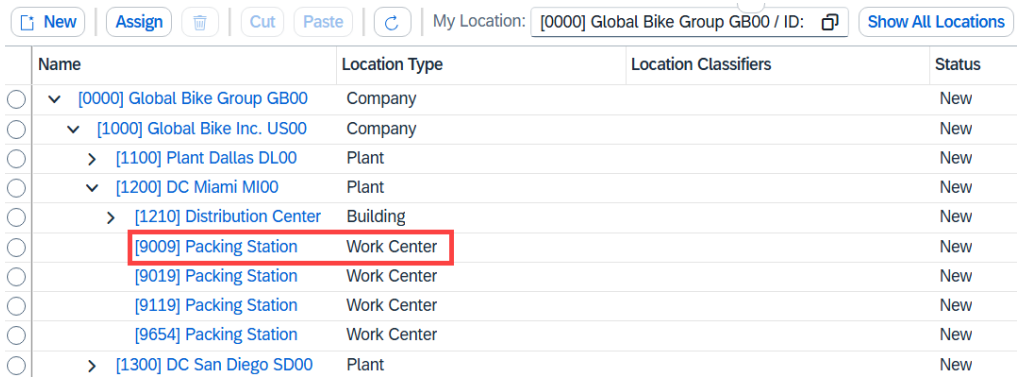
MI00
US00
BI00

On the *Adress* tab, find and select *Country/Region* **US**, *Region* **FL** and *Time Zone* **EST**.

US
FL
EST



Click Save to create the location. After that, click on Return to Location Structure. In order to find the new location you just created you need to explode the location hierarchy again.



Click on SAP to return to the SAP Fiori Launchpad.

Step 2: Add Responsible Person to Your Location

Task Add a responsible person to your new location. **Time** 5 min

Short Description After you created a new location for Global Bike in the previous step appoint Jae Kim (Plant Manager in Miami) as the responsible person for the new packing station.

Name (Position) Jae Kim, Plant Manager Miami, DC Miami

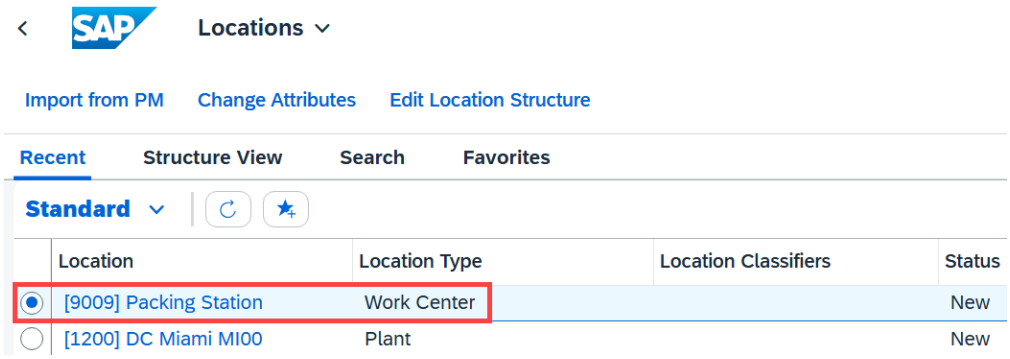
In the space *Environment Management* use the app *Manage Locations* to see all locations available to you.

Start

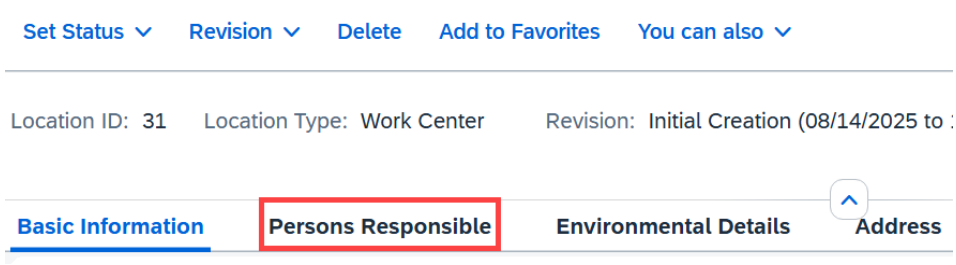


On the *Locations* screen on the *Recent* tab, you find your new location *[9###] Packing Station*. Click on the blue location name to open it.

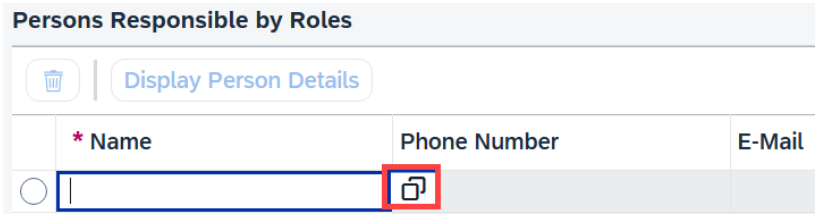
[9###] Packing Station



Then, click the *Persons Responsible* tab.

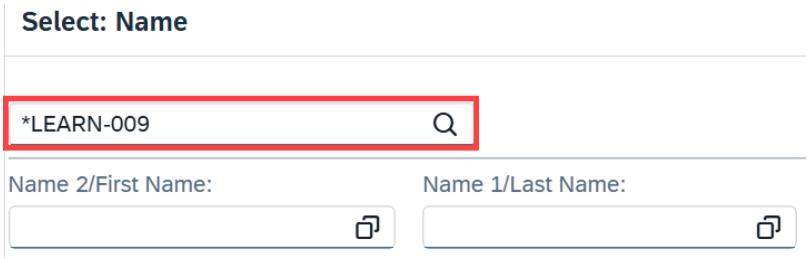


Click in the first row of the *Name* column. Then, click the value help icon that appears on the right.



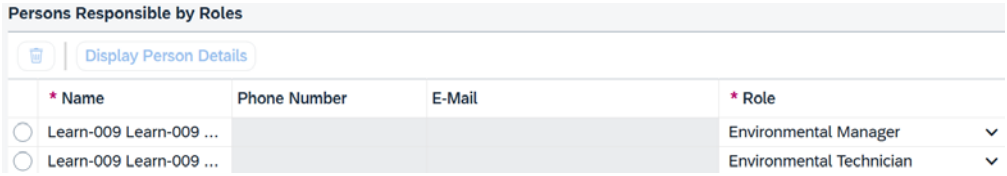
In the *Select: Name* window, in the search field type **Learn-###** (repleace the ### with your personal ID), adding an astericks sign before and after.

Learn-###



Press Go to execute the search. Click the circle left to your user name to confirm. After that, in the *Role* column of that row choose **Environmental Manager**. Repeat the same process to add a new row and to assign the role **Environmental Technican** to the same person.

Environmental Manager
Environmental Technical



Click Save to save your changes. Click on SAP to return to the SAP Fiori Launchpad.

Step 3: Create Compliance Scenario

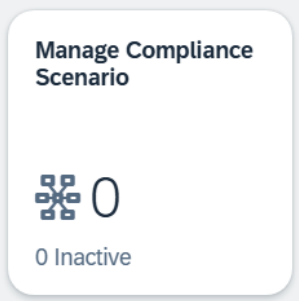
Task Create a compliance scenario for your new location. **Time** 10 min

Short Description Use the *Manage Compliance Scenario* app to create a compliance scenario.

Name (Position) Alberto Conti, Technical Office Assistant, DC Miami

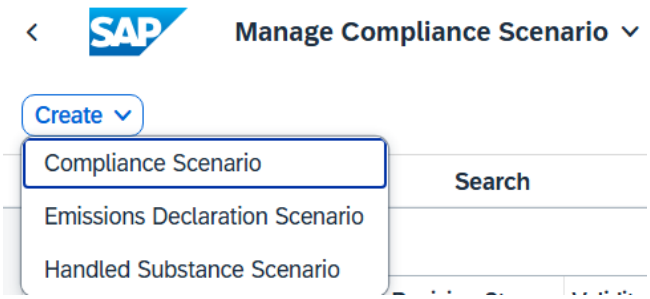
In the space *Environment Management* click on the *Manage Compliance Scenario* app.

Start



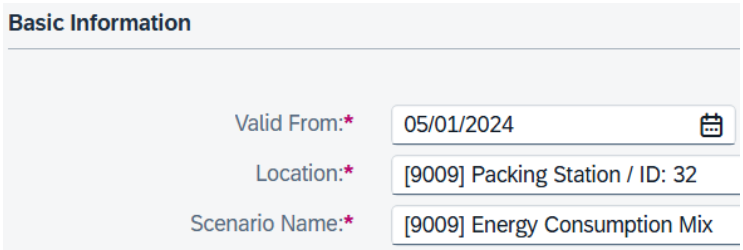
To create a new scenario, click on *Create* → *Compliance Scenario.*

Compliance Scenario



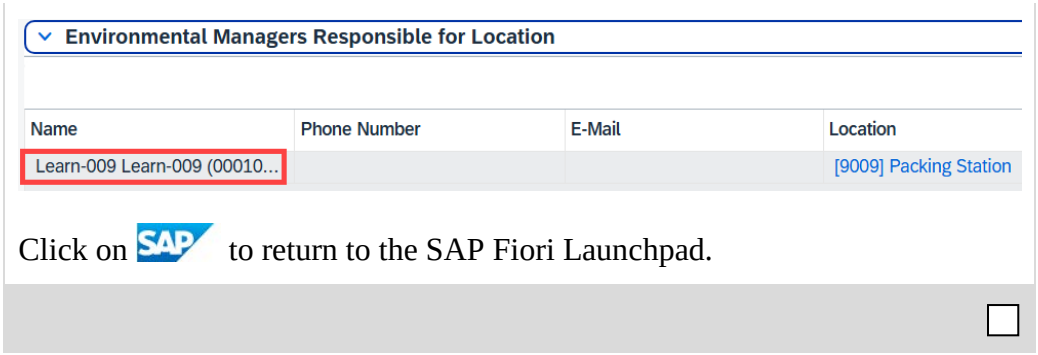
In the *Create Compliance Scenario* screen, enter **05/01/2024** (May 1, 2024) as *Valid From*. In the *Location* field, use the value help icon to search and find **[9###] Packing Station**. Then, enter **[9###] Energy Consumption Mix** as *Scenario Name*.

05/01/2024
[9###] Packing Station
[9###] Energy Consumption Mix



Compare your entries with the screenshot above. Then click Save.

In the middle of the screen, open the area *Environmental Managers Responsible for Location* to confirm that your user account Learn-### has automatically been assigned.



Click on to return to the SAP Fiori Launchpad.

Step 4: Create Data Collections

Task Create two data collections. **Time** 5 min

Short Description After creating a new compliance scenario in the previous step you can now create data collections for the energy consumption of your new packing station.

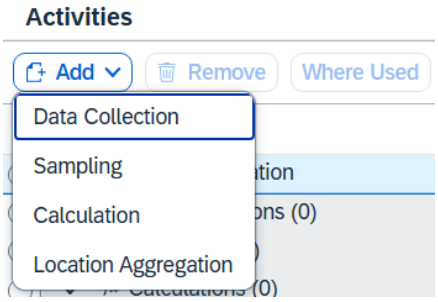
Name (Position) Alberto Conti, Technical Office Assistant, DC Miami

Open your scenario in the *Manage Compliance Scenario* app again.

Start

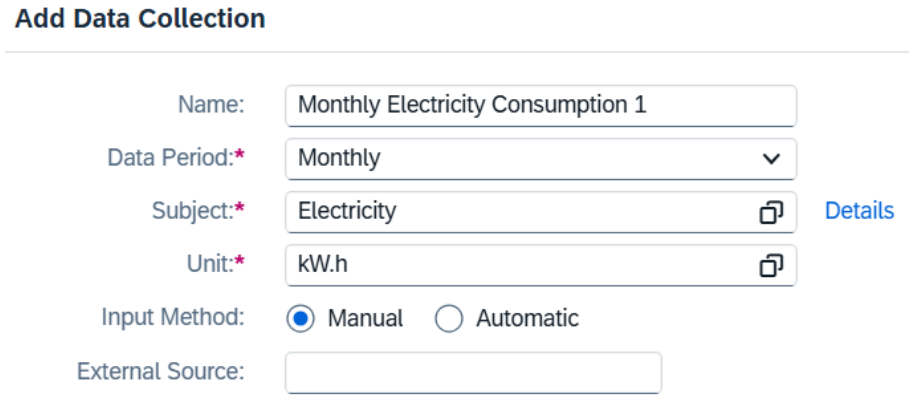
In your compliance scenario, in the *Activities* area click Add. Choose *Data Collection*.

Data Collection



In the *Add Data Collection* window, type **Monthly Electricity Consumption 1** as *Name*. For *Data Period*, select **Monthly**. In the *Subject* field, use the value help icon to find and select **Electricity** (you may have to use the *Search* tab). As *Unit*, again use the help icon and search for **kW.h** as Unit (of measure). Click OK to confirm everything.

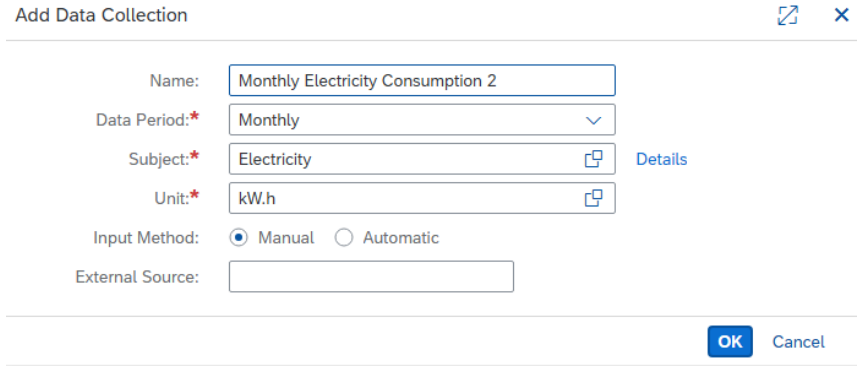
Monthly Electricity Consumption 1
Monthly
Electricity
kW.h



After that, you have to assign a *data classifier* to the collection. To do so, click on Manage Data Classifiers. Use the value help icon to find and select *Scope 1* as your classifier. A scope 1 classifier refers to consumptions resulted directly in the company. Press Assign Data Classifier to confirm. Then, click *Close*.

Scope 1

After you have created the first data collection, please create another Scope 1 collection using the same steps with the following features.



Monthly Electricity Consumption 2

Monthly

Electricity

kW.h

Don’t forget to assign data classifier *Scope 1* for this data collection as well.

Scope 1

Stay in the *Manage Compliance Scenario* app.

Step 5: Enter Data in Data Collections

Task Enter environmental data in your data collection. **Time** 15 min

Short Description Use the *Manage Compliance Scenario* app to enter data points in your new data collection.

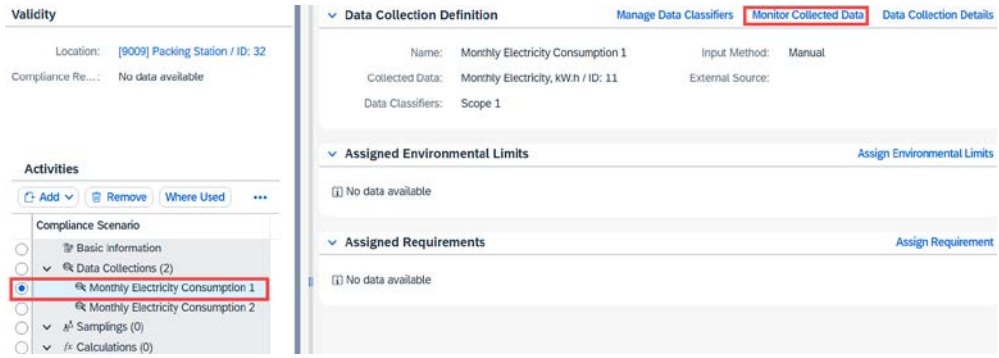
Name (Position) Alberto Conti, Technical Office Assistant, DC Miami

Your compliance scenario *[9###] Energy Consumption Mix* should still be open. If not, open your scenario in the *Manage Compliance Scenario* again.

Start

In the *Activities* area on the left, select and display your *Monthly Electricity Consumption 1* data collection. In order to enter data, in the top-right click Monitor Collected Data .

Monthly Electricity Consumption 1

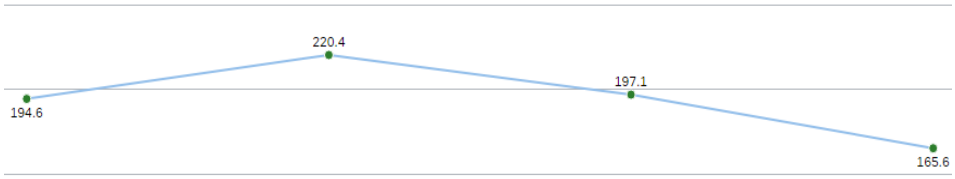


You see that no energy emission data has been logged for *Packing Station* yet. There are two possibilities to enter data. You can manually add single data points by using the Add button. Second, you can speed up the process by importing data from an Excel sheet which can include multiple data points.

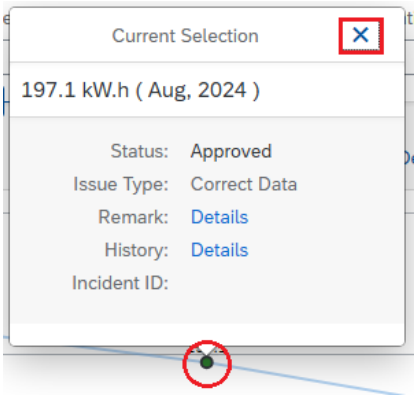
For now, you will add data points to your data collection manually. As said before, use the Add button. Enter the data points listed in the table below to your data collection.

June 2024	194.6
July 2024	220.4
August 2024	197.1
September 2024	165.6

Compare your screen to the screenshot below. You may have to adjust the time frame.

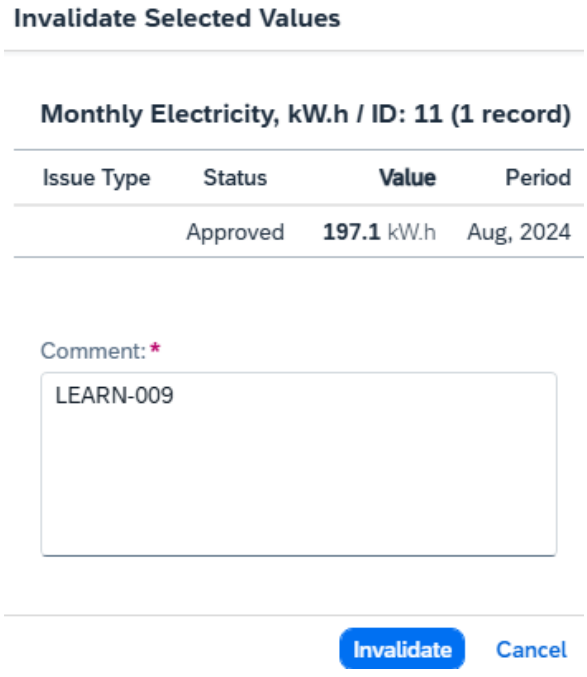


You can change the characteristics of each data points. For that, click on the *August 2024* data point. The context window that opens displays information of that data point such as its status, issues type, remarks and history. Close the window, but make sure that your data point is still selected.



After that, click Invalidate to set the status to *Not Valid*. In the opening window, in the *Comment* text box type in your user name **LEARN-###**.

LEARN-###



Click **Invalidate**. The data point will disapear from the graph. To see it again, choose every available entry in the *Status* drop-down in the filter header. Now, the data point will be shown in gray color. You may also approve it again to be a valid data point. For that, select the August data point again and click *Approve*. Use the same comment as above to confirm.

Invalidate

Approve

Close your current web browser tab *Monitor Data* and go back to your *Compliance Scenario: [9###] Energy Consumption Mix* tab.

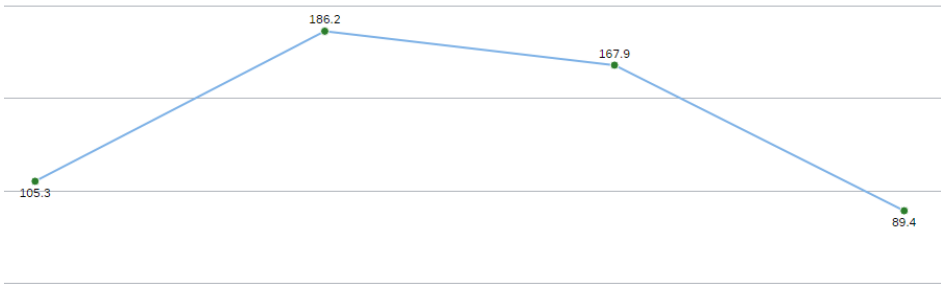


Monthly Electricity Consumption 2

Repeat the same steps for data collection *Monthly Energy Consumption 2* using the following data.

June 2024	105.3
July 2024	186.2
August 2024	167.9
September 2024	89.4

Compare your screen to the screenshot below.



Close your current web browser tab *Monitor Data* and go back to your *Compliance Scenario: [9###] Energy Consumption Mix* tab.



Stay in the *Manage Compliance Scenario* app.

Step 6: Create Calculation

Task Create a calculation for your scenario. **Time** 15 min

Short Description Use the *Manage Compliance Scenario* app to create a calculation.

Name (Position) Alberto Conti, Technical Office Assistant, DC Miami

Your compliance scenario *[9###] Energy Consumption Mix* should still be open. If not, open your scenario in the *Manage Compliance Scenario* again.

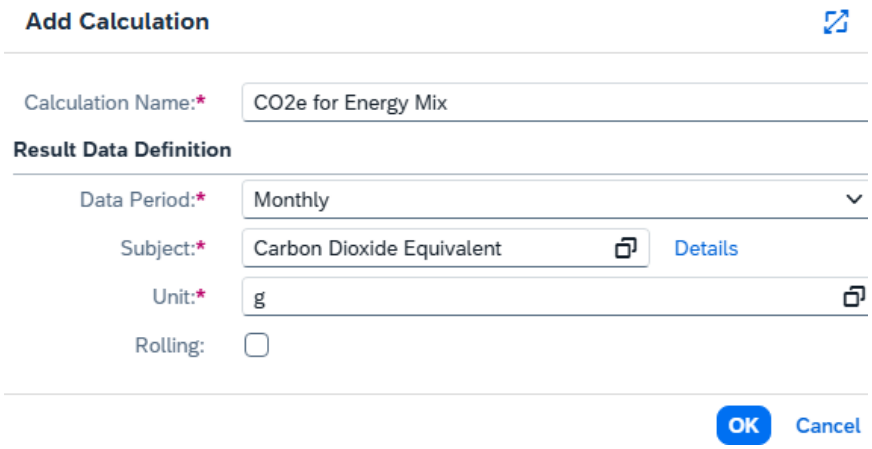
Start

In your compliance scenario, in the *Activities* area click Add and choose *Calculation*. Select the radio button *New Local Expression*.

Calculation
New Local Expression

In the *Add Calculation* window, for *Calculation Name* use **CO2e for Energy Mix**. Choose **Monthly** as *Data Period*. Use the value help icon to set **Carbon Dioxide Equivalent** as *Subject* and **g** (gram) as *Unit*. Compare your screen to the screenshot below.

CO2e for Energy Mix
Monthly
Carbon Dioxide Equivalent
g



Click *OK*. Your calculation is now created.

OK

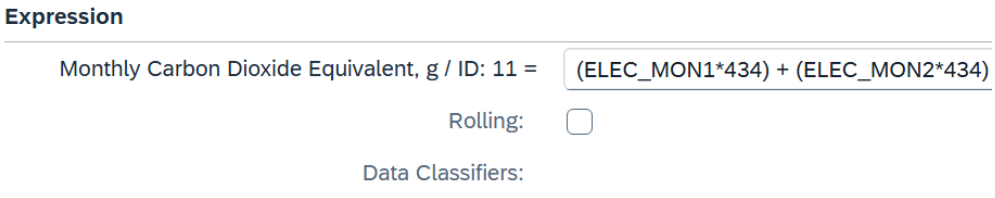
Ignore the warning message on the bottom.



Next, you have to create an expression for the calculation of carbon dioxide equivalents. To do so, click on Calculation Setup Details.

Here, you may start defining input variables for your expression. However, you may also simply type **(ELEC_MON1*434) + (ELEC_MON2*434)** in the *Expression* text box in the middle of the screen. This formula defines that your collected data points (in kW/h) will be multiplied with 434. 434 grams is the average CO2 emission conversion factor for 1 kW/h.

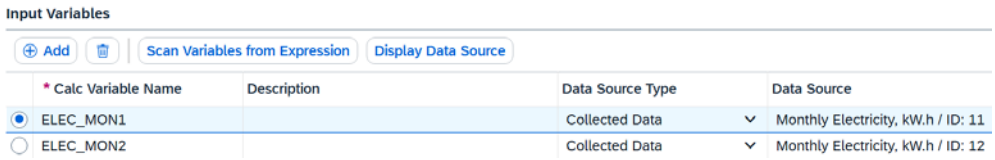
(ELEC_MON1*434) + (ELEC_MON2*434)



The variables *ELEC_MON1* and *ELEC_MON2* will later be connected to your two data collections. After that, click Scan Variables from Expression. Your variables appear in the table below.

For both variable, select **Collected Data** as *Data Source Type*. For *Data Source* use the value help icon and click Go. Select the first data collection for *ELEC_MON1* and the second for *ELEC_MON2*. Your screen should look similar to the screenshot below now.

Collected Data
Monthly Electricity Consumption 1 & 2



Click *Save*. In the top right, click Calculate Manually.

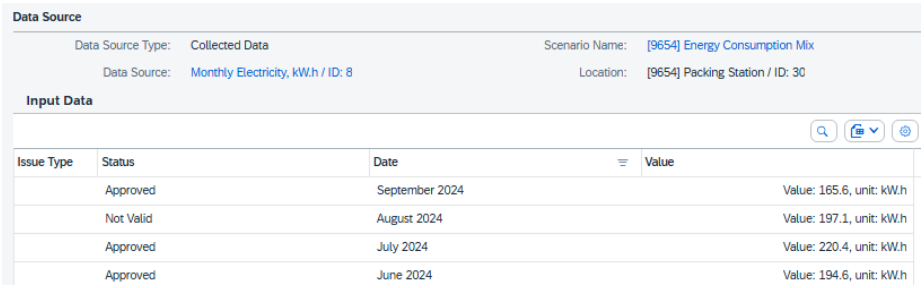
On the *Manual Calculation* screen, in the *Period* area select *From:* **June 2024** and *To:* **September 2024**. Then, click *Next*.

June 2024
September 2024

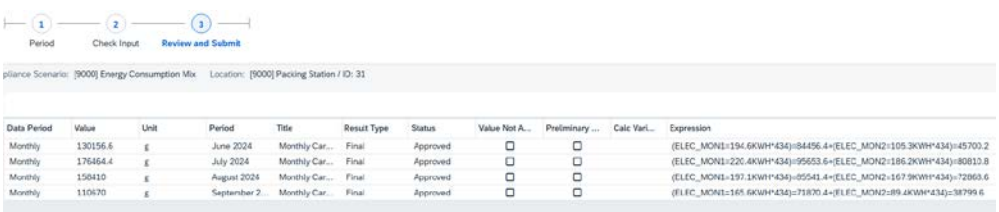


In the *Check Input* step, you may check the values for your input variables again by clicking *ELEC_MON1* and *ELEC_MON2*. Then, click *Next*.

ELEC_MON1
ELEC_MON2



In the final *Review and Submit* step, the system displays the final calculation results.



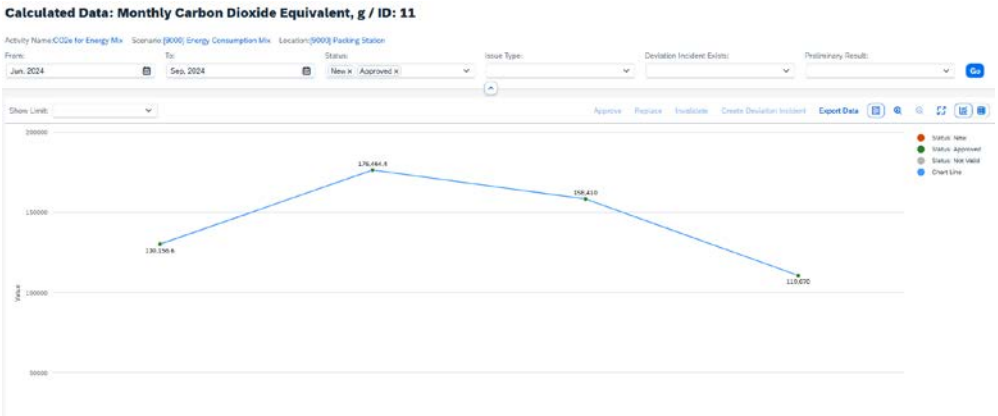
Click *Save*. Then, close your current web browser tab *Manual Calculation: CO2e for Energy Mix* and go back to your *Calculation: CO2e for Energy Mix* tab.



Back on the calculation screen, in the bottom right click *Back to Compliance Scenario*.

Back in your compliance scenario with your calculation still displayed, click Monitor Collected Data in the middle of the screen. Specify as time frame *From:* **Jun, 2024** *To:* **Sep, 2024**. The calculated data is shown in the graph below.

Jun, 2024
Sep, 2024



When you switch to *Table View*, you see that the calculated CO2 equivalents are displayed in grams.

Data Records (4)

		Issue Type	Status	Value	Period	History	Replaced	Incident ID	Responsible Person
☐		Correct Data	Approved	**110670** g	Sep, 2024	History	☐		Learn-000 Learn-000
☐		Correct Data	Approved	**158410** g	Aug, 2024	History	☐		Learn-000 Learn-000
☐		Correct Data	Approved	**176464.4** g	Jul, 2024	History	☐		Learn-000 Learn-000
☐		Correct Data	Approved	**130156.6** g	Jun, 2024	History	☐		Learn-000 Learn-000

Close your current web browser tab *Monitor Data* and go back to your *Compliance Scenario: [9###] Energy Consumption Mix* tab.



Stay in the *Manage Compliance Scenario* app.

Step 7: Create Location Aggregation

Task Create a location aggregation for your scenario. **Time** 15 min

Short Description Use the *Manage Compliance Scenario* app to create a location aggregation, which combines both data collections defined. With that feature, you can easily see the total electricity consumption of this scenario for your packing station.

Name (Position) Alberto Conti, Technical Office Assistant, DC Miami

Your compliance scenario *[9###] Energy Consumption Mix* should still be open. If not, open your scenario in the *Manage Compliance Scenario* again.

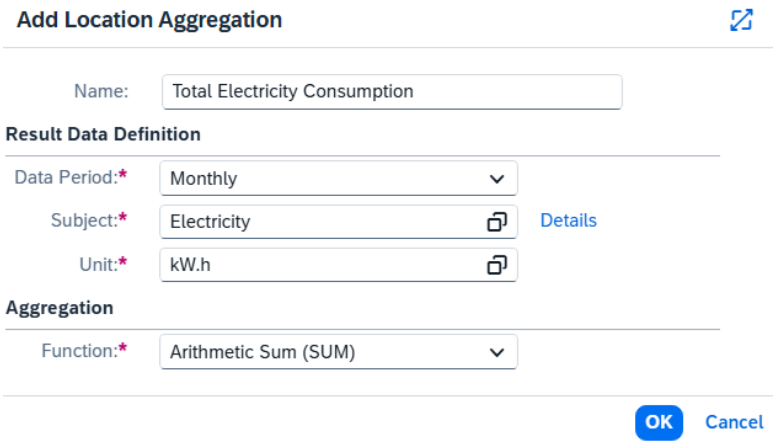
Start

In your compliance scenario, in the *Activities* area click Add. Choose *Location Aggregation*.

Location Aggregation

In the *Add Location Aggregation* window, for *Name* type in **Total Electricity Consumption**. Choose **Monthly** as *Data Period*. Use the value help icon to set **Electricity** as *Subject* and **kW.h** as *Unit*. Compare your screen with the screenshot shown below. As the aggregation function, keep *Arithmetic Sum*.

Total Electricity Consumption
Monthly
Electricity
kW.h

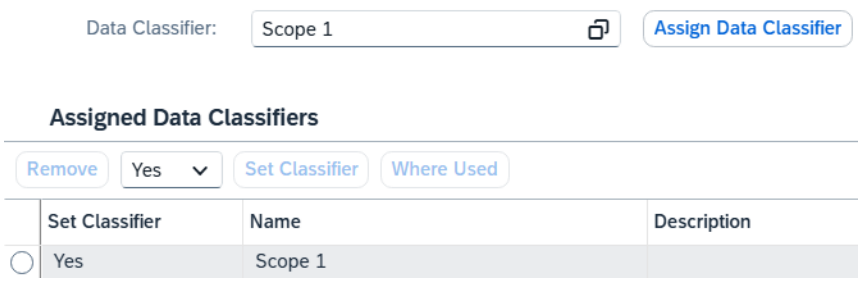


Click *OK* to create your aggregation.

OK

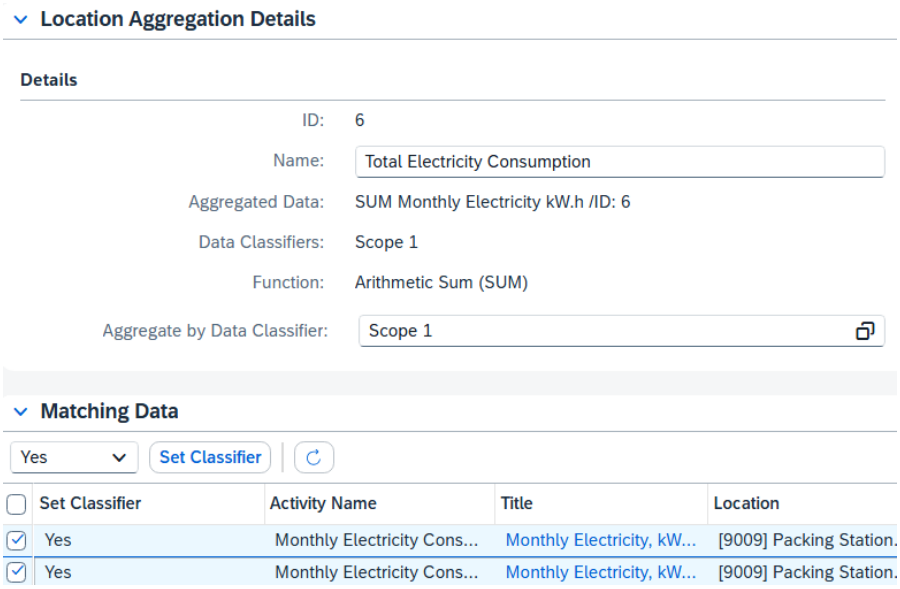
First you need to assign a data classifier to the aggregation. To do so, click Manage Data Classifiers. Use the value help icon to find and select *Scope 1*. Click Assign Data Classifier to confirm. Then, click *Close*.

Scope 1



Scope 1

You now need to link your data collections to your location aggregation. Please click Edit Aggregation Setup . In the opening web browser tab, again select **Scope 1** in the *Aggregate by Data Classifier* field. Now explode the *Matching Data* area. You will see both of your data collections automatically populated. You may have to set the classifier to **Yes**.



Jun 2024
Sep 2024

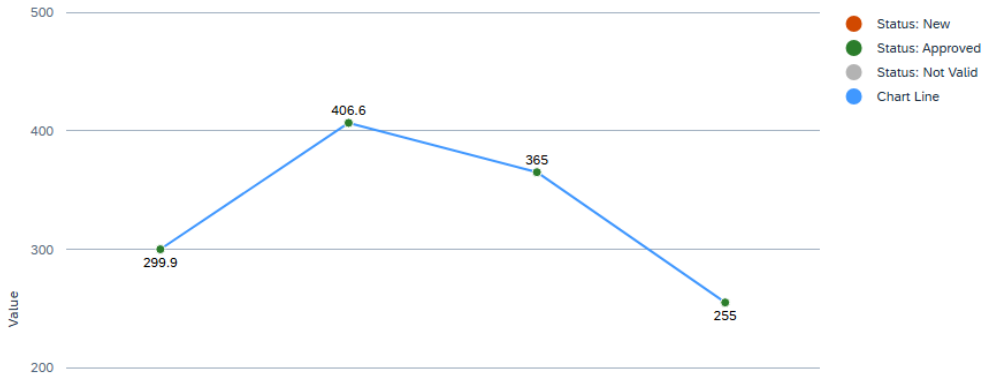
Click *Save*. Now you will aggregate the data manually. Click on Aggregate Manually . Set your *Period* from **Jun 2024** to **Sep 2024**. Click *Next* twice and *save* your manual aggregation. Your aggregation will look like this

Period	Value	Unit	Status	Value Not Availa...	Preliminary Result
June 2024	299.9	kW.h	Approved	☐	☐
July 2024	406.6	kW.h	Approved	☐	☐
August 2024	365	kW.h	Approved	☐	☐
September 2024	255	kW.h	Approved	☐	☐

Click on the Back Button to return to the aggregation screen.

Back on the aggregation screen, in the bottom right click *Back to Compliance Scenario*.

Now click on Monitor Aggregated Data . Use the same time frame as before. The data should be displayed as follows.



Close your current web browser tab *Monitor Data* to go back to your *Compliance Scenario: [9###] Energy Consumption Mix* tab.



Stay in the *Manage Compliance Scenario* app.

Step 8: Activate Compliance Scenario

Task Set your compliance scenario active. **Time** 5 min

Short Description For your compliance scenario to be fully operational you need to set its status to Active.

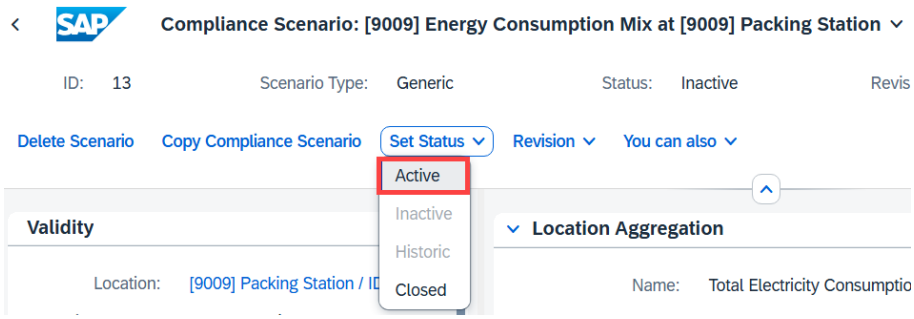
Name (Position) Jae Kim, Plant Manager Miami, DC Miami

Your compliance scenario *[9###] Energy Consumption Mix* should still be open. If not, open your scenario in the *Manage Compliance Scenario* again.

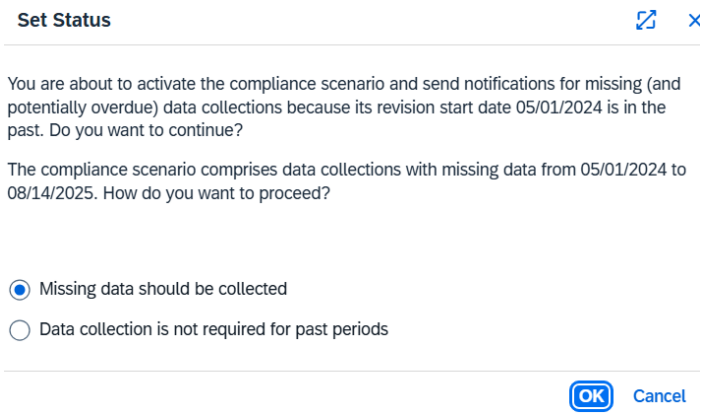
Start

On the top of your screen click *Set Status* and select **Active**.

Active



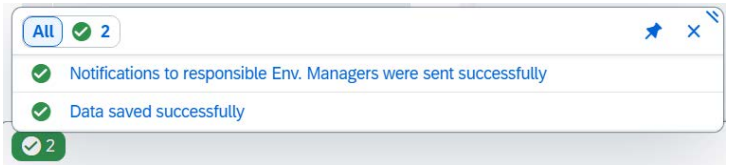
The system displays the following warning message regarding missing data.



Leave radio *Missing data should be collected* active and click *OK*.

Missing data should be collected

The system acknowledges the activation with the following success message.



Click on SAP to return to the SAP Fiori Launchpad.

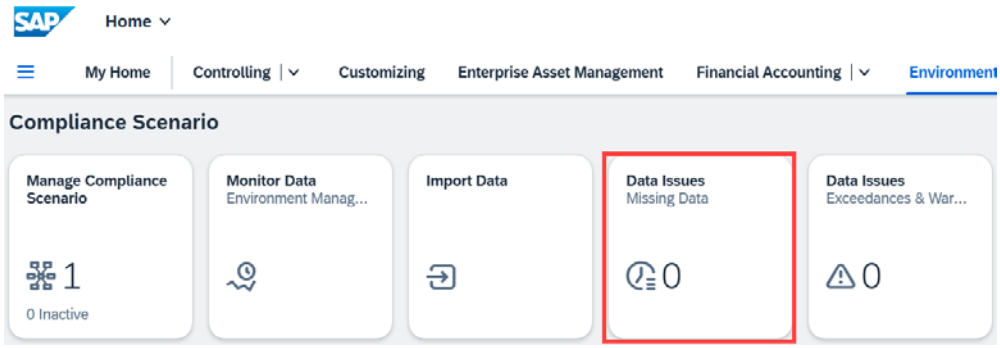
Step 9: Handle Data Issues

Task Handle data issues in your compliance scenario. **Time** 10 min

Short Description Use the *Data Issues* app to check and handle data issues resulting from missing data in your compliance scenario. Although your scenario has its revision start date in May 2024 you have not yet entered data for that month. As the environmental manager for this location you can replace missing data with regular data points.

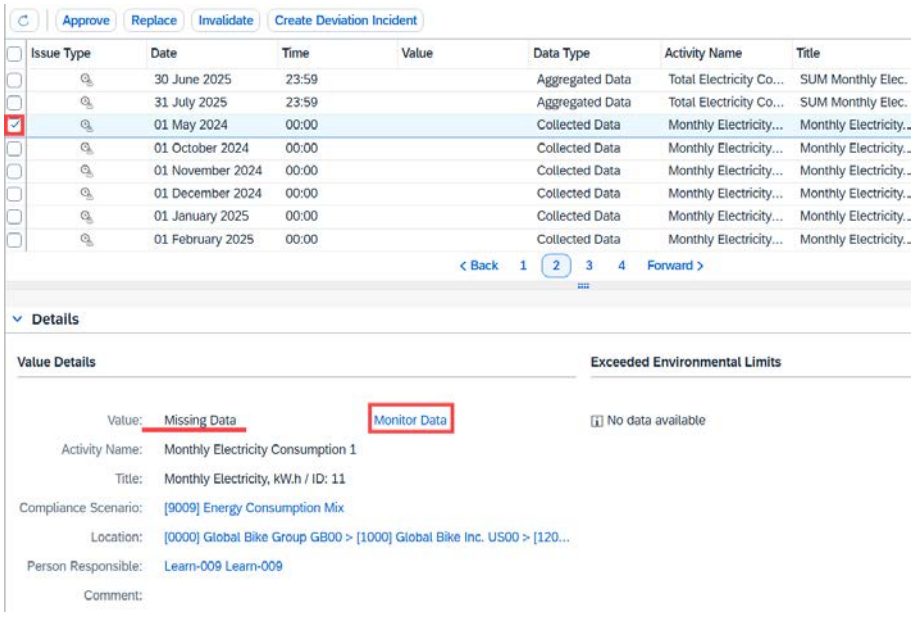
Name (Position) Jae Kim, Plant Manager Miami, DC Miami

In the *Environment Management* space, in the *Compliance Scenario* section click on the *Data Issues* app. Start

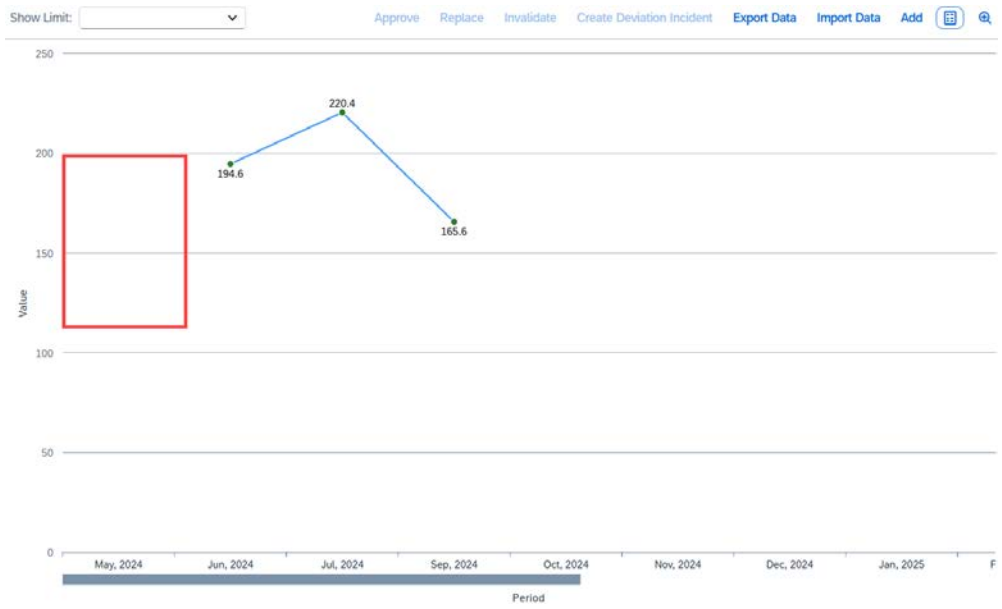


Note Depending on the month and year in which you are processing this case study you may see more missing data issues for future periods.

In the *Data Issues* app, you should find multiple missing data points for your location. Among them are two entries for May 2024. Check the details for the issue with your data collection *Monthly Energy Consumption 1* by selecting the first entry. To identify the missing data, click *Monitor Data*. Monitor Data



You see that the data point for May 2024 is missing.

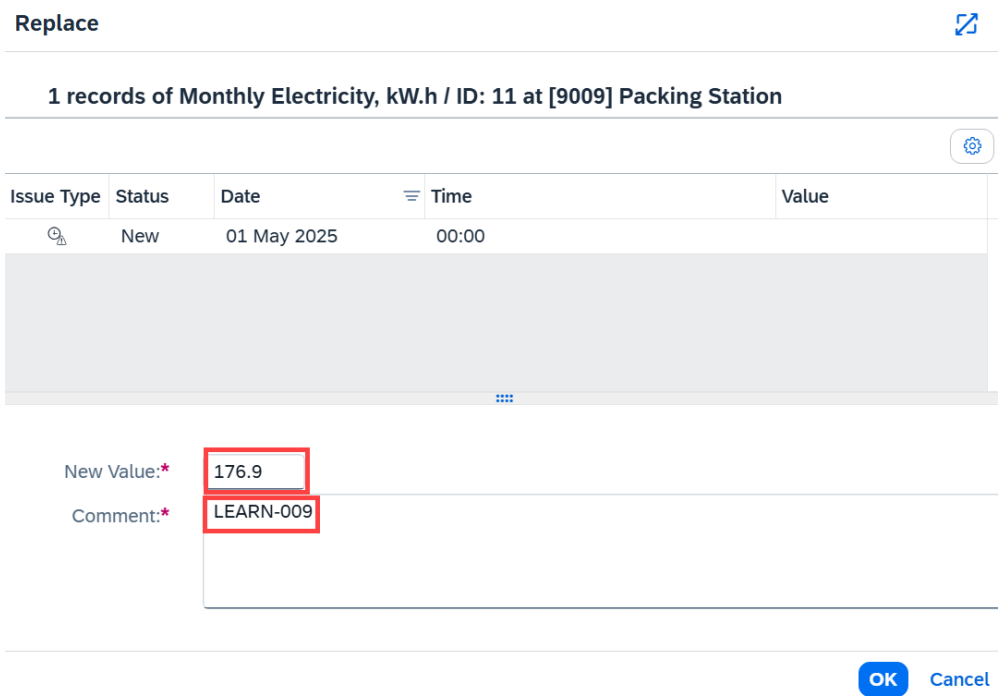


Close your current web browser tab *Monitor Data* to go back to your *Compliance Scenario: [9###] Energy Consumption Mix* tab.



176.9
LEARN-###

Back in the Data Issues app, make sure that your data collection *Monthly Energy Consumption* 1 is still selected. Then, click the Replace button. In the following window, enter **176.9** as *New Value* and your user account **LEARN-###** as *Comment*. Confirm with *OK*.



As an immediate result you see that the first data issue disappeared. However, two new data issues occurred. Please check the new issues.

Question Which activities are causing the new issues and why?

Answer _____

_____

_____

_____

_____

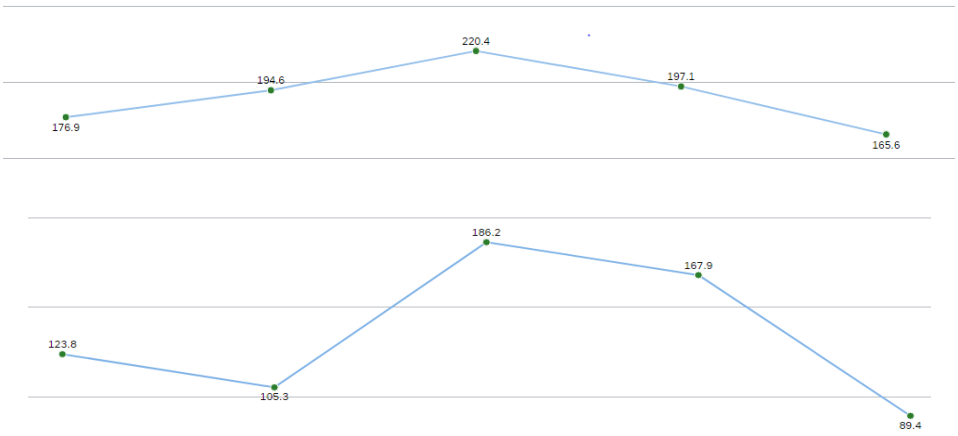
Before dealing with the new issues, you take care of the missing data in your collection *Monthly Energy Consumption 2*. Follow the exact same steps as described above and enter a May 2024 data point of **123.8** for your second collection.

123.8
LEARN-###



Click on SAP to return to the SAP Fiori launchpad.

Open your compliance scenario in the *Manage Compliance Scenario* app to check the new data points in your collections for May 2024.



Click on SAP to return to the SAP Fiori launchpad.

Step 10: Prepare Environmental Reports

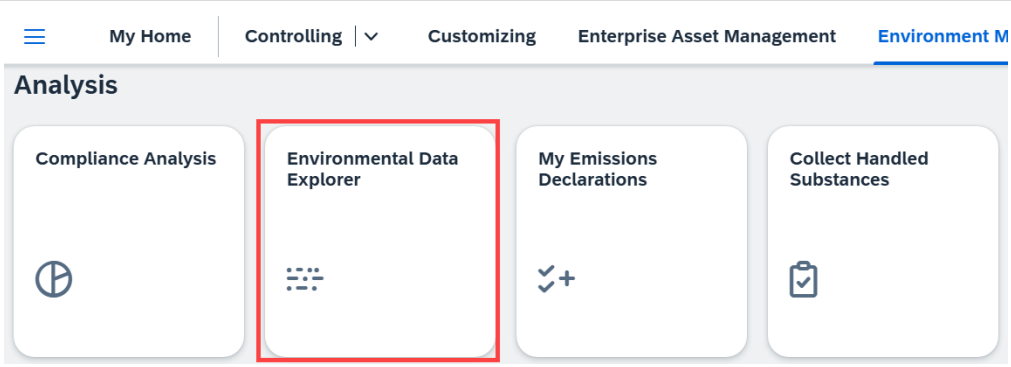
Task Prepare environmental data for ESG reports. **Time** 10 min

Short Description Use the *Environmental Data Explorer* app to prepare and export environmental data for your new compliance scenario.

Name (Position) Alberto Conti, Technical Office Assistant, DC Miami

Start

In the *Environment Management* space, in the *Analysis* section click on the *Environmental Data Explorer* app.

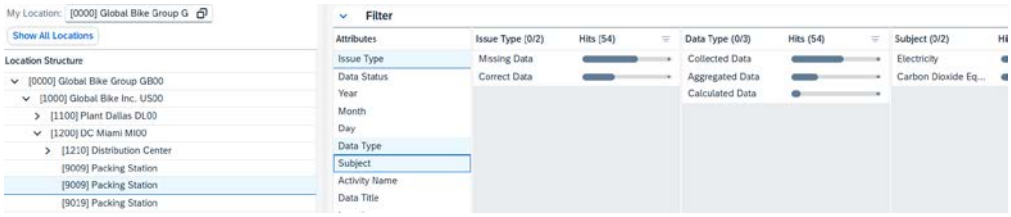


Note: You must first click on the name *[1000] Global Bike Inc. US00* before the tree can be opened. Only then can the tree be opened or expanded.

[9###] Packing Station

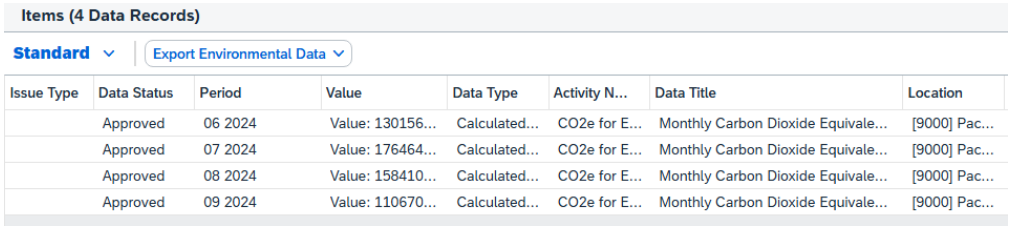
In the following window you see the location structure on the left side and its environmental data in the middle. Expand the hirarchical location structure to find your location **[9###] Packing Station** and click on the entry to select it.

At this point, you could already export all data records found. However, you can further analyze and restrict the the environmental data set for your location based on predefined filter attributes. In your case, select *Issue Type*, *Data Type* and *Subject* as relevant filters.



Carbon Dioxide Equivalent
Correct Data

For an ESG report of the monthly carbon footprint of the new packing station exclude the missing data by clicking on **Carbon Dioxide Equivalent** and **Correct Data**. This results in four instead of five data records.



Question Can you explain which CO2e data record is missing and why?

Answer _____

_____

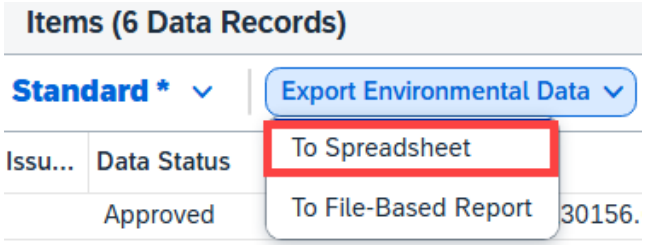
_____

_____

_____

To download your analysis result, click *Export Environmental Data* and select *To Spreadsheet.*

Export Environmental Data
To Spreadsheet



In this simple example, the Excel sheet comprises the four data records selected in and exported from the *Environmental Data Explorer* app.

Task Use the *Environmental Data Explorer* app to filter and export all existing data for electricity consumption in your *[9###] Packing Station*.

Click on SAP to return to the SAP Fiori launchpad.

Challenge 1: Solve Missing Data Errors

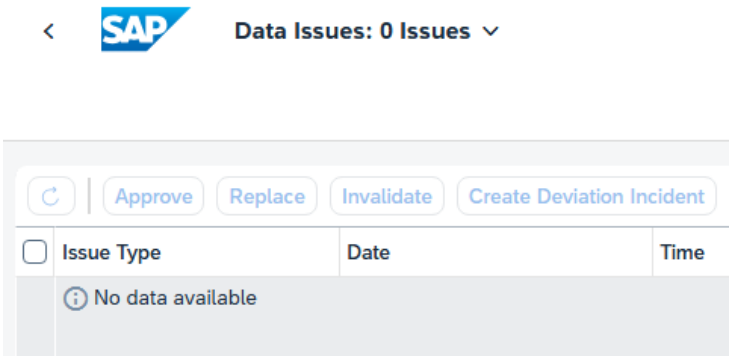
Challenge Solve all remaining missing data errors. **Time** 20 min

Description Use the *Data Issues* app to check and handle all remaining data issues resulting from missing data in your compliance scenario.

Scenario

Remember that you had solved two missing data errors in a previous step. In return, however, the new data points for May 2024 in both of your collections had caused two new missing data errors. Analyze their root cause and solve them. All information and practical knowledge necessary for this challenge is given in this case study.

You will have succeeded with this challenge when there are no more missing data issues for the reporting period.



Challenge 2: Calculate Annual CO2 Equivalent

Challenge Create another data calculation. **Time** 20 min

Description Use the *Manage Compliance Scenario* app to calculate the annual CO2 equivalent for the new packing station.

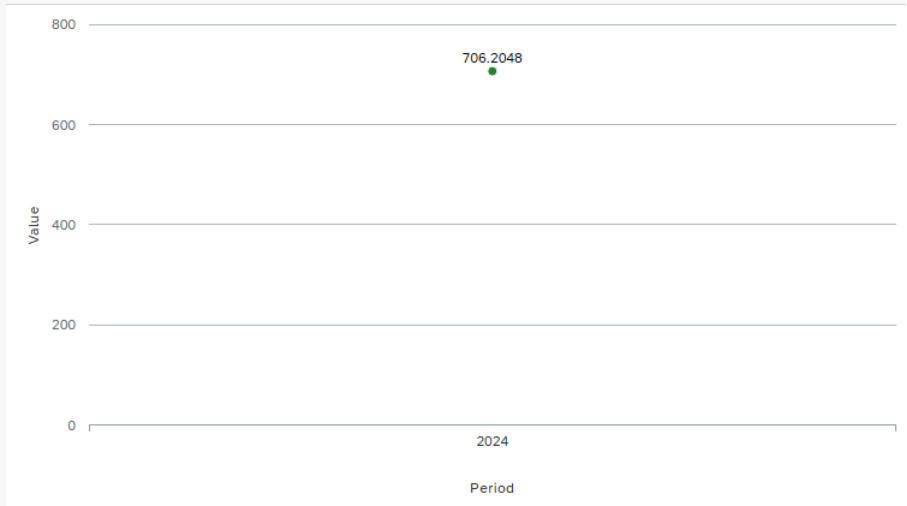
Scenario

For most ESG reports, environmental KPIs are summarized to annual figures. You are tasked to create the KPI *Annual CO2e for Energy Mix*.

Requirement 1 Due to the size of this value express your result in kilograms, not grams.

Requirement 2 Present your result in a chart and in a table.

You will have succeeded with this challenge when you are able to show the figures shown below for 2024.



Data Record (1)

		Issue Type	Status	Value	Period	History	Replaced	Incident ID	Responsible Person
☐		Correct Data	Approved	706.2048 kg	2024	History	☐		Learn-000 Learn-000